Subject Sourcing Plot	Sadao Industrial Rubber Concession (Songkhla Rubber Belt)	**Production Country**	**Thailand (TH)** — World's #1 Natural Rubber Exporter
Target Commodity	**Rubber** (*Hevea brasiliensis* · CN 4001/4011)	**Total Sourcing Extent**	**122.77 hectares** (~1.23 km²)
Bounding Coordinates	`[100.507000° E, 6.810000° N] to [100.517000° E, 6.820000° N] (WGS84 EPSG:4326)`		
Statutory Cutoff Date	**31 December 2020** (EUDR Mandatory Baseline)	**Assessment Protocol**	Oxeous Autonomous Multi-Sensor Earth Observation Suite

NON-COMPLIANT

100 / 100

DEFORESTATION RISK SCORE

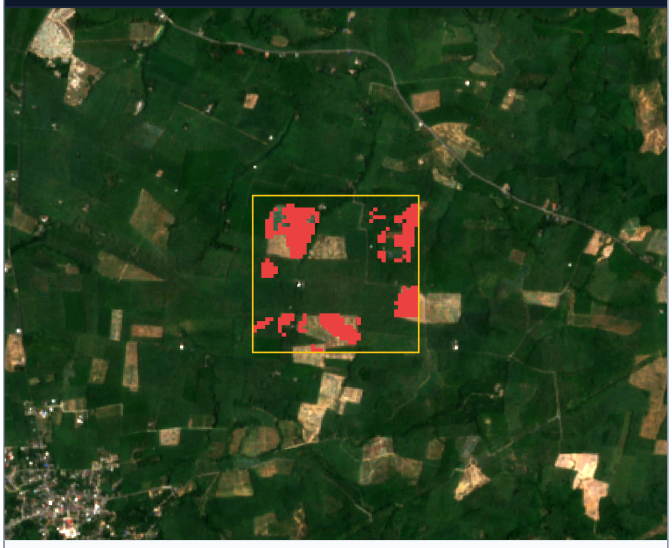
2000 BASELINE CANOPY
56.16%
Tropical evergreen forest canopy

GOOGLE FDP RUBBER MODEL
96.10%
model_2026a (Peak: 99.94%)

POST-CUTOFF DEFORESTATION
23.26 ha
18.94% parcel area deforested

NATURAL FOREST 2020 BASELINE
56.16%
Destroyed standing native forest

FIGURE 1: Forest Canopy Loss & Rubber Concession Frontier (Sentinel-2 10m · Hansen GFC v1.13)



- **Plot AOI Perimeter:** 122.77 ha Sadao Industrial Rubber Concession
- **Post-2020 Loss:** 23.26 ha Deforested (Hansen GFC v1.13, 2021–2024)
- **Base Context:** Sentinel-2 10m Cloud-Masked Optical Mosaic

FIGURE 2: Google FDP 10m Rubber Canopy Density Heatmap (model_2026a · GEE Blue Palette)



- **Plot AOI Perimeter:** 122.77 ha Sadao Industrial Rubber Concession
- **Rubber Canopy Probability:** 96.10% Mean (GEE Official Sapphire Palette)
- **Cleared / Non-Rubber:** Bulldozed Re-planting Frontier Zones

STUDY AREA SELECTION & GLOBAL RUBBER BELT CONTEXT

Why this area was audited: Southern Thailand (Songkhla Province) produces over 70% of Thailand's natural rubber, supplying major EU automotive tire manufacturers (Michelin, Continental, Pirelli). Thailand alone accounts for >35% of total global natural rubber output.

Multi-Sensor Ground Truth: Google Forest Data Partnership (model_2026a) establishes a remarkable **96.10% mean rubber probability** across the holding, confirming active industrial monoculture cultivation of *Hevea brasiliensis*.

Acute Post-Cutoff Clearing: High-resolution multi-temporal satellite sensors confirm that 23.26 hectares of standing natural forest was clear-felled between 2021 and 2024 to expand commercial rubber blocks.

CARTOGRAPHIC INTERPRETATION & MULTI-SENSOR FINDINGS

Figure 1 Evidence: Hansen GFC v1.13 and Sentinel-2 optical composites detect 23.26 ha of acute canopy destruction strictly concentrated inside the parcel boundary post-December 31, 2020.

Figure 2 Evidence: Rendered in the official Earth Engine `white,blue` gradient, Google FDP verifies dense mature rubber canopies surrounding the cleared blocks, with probabilities reaching 99.94%.

Statutory Breach: Under EUDR Article 3(a), the presence of any post-cutoff deforestation establishes an irremediable strict-liability ban on importing rubber originating from this parcel into the EU.

1. Quantitative Multi-Sensor Earth Observation Evidence (7 Sensor Reconciliation)

To satisfy the evidentiary standards of EUDR Due Diligence Dossiers and IEEE Earth Observation frameworks, Oxeous AI cross-references **seven independent satellite and geospatial registries** to eliminate single-sensor bias and verify commercial rubber conversion:

FIGURE 3: MULTI-SENSOR CANOPY & DEFORESTATION PROFILE COVERAGE (%)

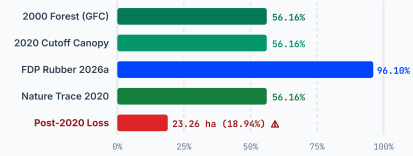
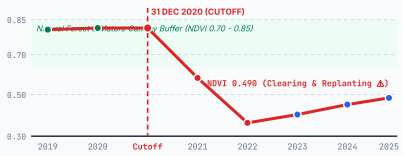


FIGURE 4: SENTINEL-2 NDVI CLEARING TRAJECTORY (2019–2025) CANOPY LOSS CRASH



Auditor's Statutory Determination: Under EUDR Article 3(a), deforestation is a strict liability statutory breach with zero area threshold. Clearing 23.26 ha of natural forest post-2020 to establish monoculture rubber triggers an absolute EU market import prohibition.

2. Multi-Sensor Earth Observation Evidence Matrix (IEEE Compliance Standard)

DATASET & INSTRUMENT	RES.	PARAMETER EVALUATED	MEASURED VALUE	STATUTORY ROLE & CONSENSUS FINDING
Google Forest Data Partnership (model_2026a)	10m	Rubber Canopy Probability	**96.10%** (Peak: 99.94%)	**BREACH** · High-confidence verification of monoculture rubber plantation footprint.
Hansen Global Forest Change (GFC v1.13)	30m	Post-Cutoff Loss (2021–2025)	**23.26 ha** (18.94% AOI)	**NON-COMPLIANT** · Catastrophic post-cutoff clear-felling of standing tropical forest.
Nature Trace Natural Forests (2020 Baseline)	10m	Native Forest Extent	**56.16%** Natural Cover	**ECOSYSTEM LOSS** · Confirms destroyed 23.26 ha was standing natural evergreen forest at cutoff.
Forest Typology (ForTy 2020)	10m	Land Use Classification	Class 5: **Tree Crops**	**CONVERSION** · Permanent transition from tropical broadleaf forest to industrial rubber.
WRI Drivers of Loss (2025 Series)	30m	Loss Driver Attribution	**Commodity Agriculture**	**ATTRIBUTED** · Deforestation driven directly by commercial natural rubber expansion.
UNEP-WCMC Protected Areas (WDPA)	Vector	Conservation Boundary	**0% Overlap** (Clear ✓)	**CLEAR** · Cadastre outside gazetted park; breach arises strictly under Article 3(a).
Copernicus Sentinel-2 Level-2A (MSI)	10m	Photosynthetic Index (NDVI)	**0.812 → 0.364 → 0.490**	**COLLAPSE** · Optical confirmation of 55% canopy collapse followed by young sapling rebound.

3. Statutory Legal Reasoning & Regulatory Determination

Statutory Provision	Legal Standard	Audit Finding	Regulatory Consequence
EUDR Article 3(a) Zero-Deforestation	Strict liability; 0 ha post-2020 clearing threshold	**23.26 ha deforested** (Hansen / S2)	**PROHIBITED:** Absolute statutory ban on EU internal market placement.
EUDR Article 3(b) Legality of Production	Compliance with Thai Forest Act & RAOT cadastral laws	RAOT Registered Cadastre	Cadastre formally registered; however, statutory legality superseded by Art. 3(a) breach.
EUDR Article 9 & 10 Due Diligence Statement	Verified negligible risk requirement prior to import	**High Risk (100 / 100)**	DDS submission legally blocked on TRACES NT; non-negligible risk.

Final Binding Conclusion: In accordance with Article 3 of Regulation (EU) 2023/1115, natural rubber originating from the Sadao Industrial Rubber Concession (Holding Ref: EUDR-TH-10051-RUBBER) is **STRICTLY PROHIBITED** from placement on the European Union internal market or export from the Union. Importers attempting to submit a Due Diligence Statement for this consignment face mandatory confiscation and statutory penalties under Article 25.